DIVISHA SINGH

AI/ML | Full-Stack Development

Gmail · LinkedIn · GitHub · Portfolio

EDUCATION

B.Tech Computer Science & Engineering - Specialization Artificial Intelligence
Gautam Buddha University | 2023 - 2027

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C, R, HTML, CSS,
Full-Stack Development: Django, React, FastAPI, Tailwind CSS
AI / ML: Scikit-learn, XGBoost, NLP, Sentence Transformers
Generative AI: RAG, LLMs, Embeddings, Vector Search, Prompt Engineering
Databases & Tools: MySQL, ChromaDB, PostgreSQL, SQLite, Git, GitHub, Vite, MLflow

EXPERIENCE

Intern - IIIT-Allahabad | Semester Internship | August 2026 - Present

- Currently undertaking a technical internship focused on Machine Learning.
- Building practical experience through guided tasks, technical research, and hands-on learning.

Summer Intern - IIIT-Allahabad | July 2026 - August 2026

- Implemented a full Explainable AI layer (SHAP, Anchors, DiCE counterfactuals) to interpret model predictions at global and individual levels, surfacing actionable findings including a threshold effect in income-based risk assessment and an over-reliance on a single repayment-progress heuristic covering 75% of 'safe' predictions.
- Skills : ML, Explainable AI, Debugging, Data Wrangling, Experiment Tracking, Analytical Reasoning.

PROJECTS

Diabetes RAG System / Python • RAG • ChromaDB • FastAPI • React

- Built a RAG system that synthesizes evidence from 384 PubMed research papers to answer comparative questions about Type 2 diabetes drug therapies.
- Implemented semantic retrieval using Sentence Transformers and ChromaDB, with an LLM-based response generation pipeline using Groq.
- Designed citation-backed responses with abstention when retrieved evidence does not support a claim, reducing unsupported/ hallucinated answers.
- Evaluated the system using an LLM-as-judge approach, achieving 4.38/5 faithfulness and 4.12/5 answer relevance.

Loan Default Prediction with Explainable AI / Python • XGBoost • Scikit-learn • MLflow • Deepchecks

- Developed classification models for loan default prediction XGBoost and Scikit-learn.
- Addressed class imbalance using SMOTE and class-weighting techniques.
- Tracked and compared machine-learning experiments using MLflow.
- Applied explainability techniques including SHAP, Anchors, and DiCE to interpret model predictions.

AI Resume Screener / Django • Tailwind CSS • NLP • Sentence Transformers

- Built an AI-powered application to compare resumes against job descriptions and generate a matching score.
- Implemented semantic similarity and skill matching using Sentence Transformers.
- Added resume information extraction and candidate matching score generation and displayed missing skills and AI-generated recommendations.
- Displayed missing skills and AI-generated recommendations to help candidates identify areas for improvement.

CERTIFICATIONS

Artificial Intelligence & Robotics Trainee

Center for Intelligent Robotics, IIIT Allahabad | Jul 2025 - Aug 2025

- Completed an intensive training program covering AI fundamentals, machine learning, intelligent systems, robotics, and practical AI-driven applications.